Read the Room: Supplementary Material

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TABLE I

PER-ROBOT ACQUISITION REWARDS IN THE SEPARATE PPO COMPARISON. RATES ARE MULTIPLIED BY $\Delta t = 0.02\,\mathrm{s}$; EVENT REWARDS ARE ADDED ONCE. THE FROZEN FULL-SYSTEM ACTOR INSTEAD MINIMIZES IMITATION LOSS.

Term	Rate / event value
Approach progress	$6\,\mathrm{clip}((d_{t-1} - d_t)/\Delta t, -0.1, 0.1)$
Proximity	$2\exp[-(d_t/0.10)^2]$
Alignment, original	$\exp[-(\alpha_t/0.35)^2]$
Alignment, gated	$\exp[-(\alpha_t/0.35)^2 - (d_t/0.15)^2]$
Lift height	$10H(z_t - z^*)$
Lift levelness	$3\exp[-\lVert\eta_t\rVert^2/0.15^2]$
Approach / lift costs	$-0.5 - 4(\beta_i/\beta_{\max})^2$
Transition / failure	$+2$ each at both-attached and lift-complete; -5 on physical termination.

I. SEPARATE PICKUP PPO EXPERIMENT

This experiment is not an update to the frozen full-mission actor. It compares pickup reward variants, not end-to-end transport performance. Two separate PPO branches each run 65,536 steps with an asymmetric 181-input critic [1], [2]. They use generalized advantages [3], a 0.3 KL anchor, value coefficient 0.5 and zero entropy coefficient. Table I defines their acquisition rewards; the variants differ only in distance gating of orientation reward. Transport rewards retain common command-tracking, posture, grasp and regularization terms, detailed in the accompanying reward specification. All evaluations use deterministic actor means.

For maximum hand-position/orientation errors d_t, α_t, object roll/pitch η_t and base inclination β_i, the normalized height term is

$$\Phi(e) = \left(e^{-(e/0.08)^2} + 0.5e^{-(e/0.25)^2}\right)/1.5,$$
$$H(e) = \max\left\{0, \frac{\Phi(e) - \Phi(e_0)}{1 - \Phi(e_0)}\right\}, \tag{1}$$

where $e = z - z^*$ and $e_0 = z_{\text{rest}} - z^*$. At zero initial error, $H = \Phi$. The inclination scale is $\beta_{\max} = 2°$.

Both branches use one environment, horizon 128 and 512 updates; the learning rate is 3×10^{-5}, with three epochs, four minibatches, clip ratio 0.2, $\gamma = 0.99$ and GAE $\lambda = 0.95$. Normalization is frozen and the critic receives 32 warmup updates. The network has ELU hidden layers 512–256–128. Mass is sampled from $U[0.75, 3]$ kg, dimensions from $U[0.9, 1.1]$, base offsets from 0.08–0.65 m, headings within $\pm 30°$, and activation delays within 0–2 s. COM perturbations are $\pm(0.05, 0.12, 0.05)$ m and servo-gain factors 0.85–1.15. Automatic curricula are disabled.

Frozen pickup outcomes are 40/40 for the original reward and 39/40 for the distance-gated reward. Both stochastic training branches finish at 21/37 completed episodes. There is no demonstrated improvement from reward gating, and neither result establishes full-mission reliability.

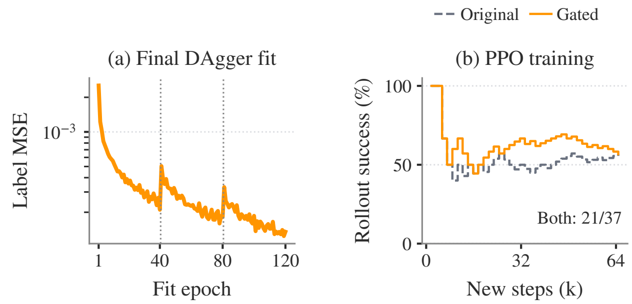

Fig. 1. Archived imitation and separate pickup PPO training traces. The two panels correspond to different learning procedures and evaluations.

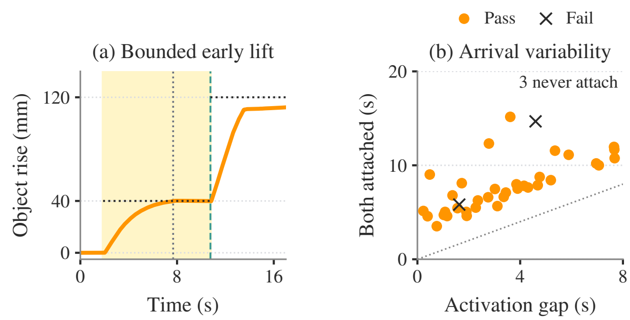

Fig. 2. Independent arrival. (a) Loading dock, seed 142013: measured object rise under a 7.68 s activation gap. Shading marks early lift; vertical dotted/dashed lines mark late activation/second attachment; horizontal lines mark height targets. (b) Both-attachment times for the 37 trials that acquire the object; color marks final mission outcome. The other three trials never attach both robots. The diagonal is $t = \Delta$.

II. ASYNCHRONOUS ACQUISITION TRACE

III. PLANAR IMITATION PROTOCOL

The mission actor is trained by imitation and DAgger [4]. A geometric teacher labels approach/lift, and a velocity actor labels transport. Three aggregation rounds collect 30 episodes each: teacher execution first, then student execution with teacher labels and no interventions. Stable-hold collection succeeds in 86/90 episodes. The final dataset contains 524,490 local observations, including 345,484 imported samples. Phase-balanced weighted action MSE uses the settings in Table II and frozen normalization. The mission actor receives no subsequent PPO update.

Imported data comprise 227,902 samples from the earlier all-object actor and 117,582 from the independent-arrival actor, totaling 65.9% of the final dataset. In 22,205 early-lift

TABLE II

COMPACT LEARNING AND EXECUTION SETTINGS.

Component	Configuration
Actor input / output	157 observations; 21 actions per robot.
Network	ELU MLP, 512–256–128 hidden units.
Physics / policy	200 / 50 Hz; identical actor weights.
Imitation	Adam 3×10^{-4}; 3 fits, 40 epochs each.

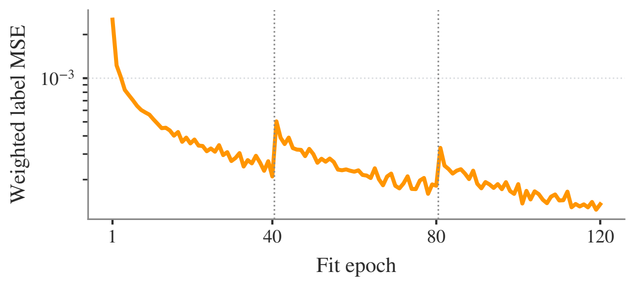

Fig. 3. Imitation fitting across three 40-epoch rounds; dotted lines mark dataset aggregation. Collection succeeds in 30/30 teacher episodes, then 28/30 and 28/30 student episodes. This is training evidence, not a matched comparison with the teacher at full-mission evaluation.

rows, the height-reference feature was rebased from 120 to 40 mm; historical action labels, other features and observation noise were retained. These are reused development data, not an independent validation set. The initialization and dataset hashes are retained with the supplementary provenance record.

The transport teacher is the saved shape-conditioned velocity actor `velocity_v6_shapes`. In furnished scenes, collection commands zero transport velocity after lifting; navigation is supplied by RRT/MPC at test time. Thus successful collection is not evidence of learning complete doorway missions from demonstrations. The sampler balances the nonempty approach, lift and transport groups; the implementation minimizes

$$\mathcal{L} = \frac{1}{|B|} \sum_{j \in B} \frac{1}{21} \sum_{k=1}^{21} w_k \left(\pi_\theta(\bar{o}_j)_k - a^*_{jk}\right)^2. \tag{2}$$

Here $\bar{o}$ uses frozen observation normalization. The bootstrap script's recorded defaults are batch size 512 and weights $w = (4, 4, 2, 4, 0.5, 1, \ldots, 1)$; equal phase allocation yields 510 samples when all three groups are present. The saved run metadata do not independently record every fitting CLI override, limiting exact reconstruction from metadata alone. Configuration and implementation details accompany the manuscript.

DAgger collection randomizes mass $U[1, 2]$ kg, dimension factors $U[0.95, 1.05]$, rail friction $U[0.9, 1.8]$, base offsets 0.08–0.50 m, headings $\pm 20°$ and activation delays 0–8 s across ten scenes. Automatic stage/difficulty curricula are disabled; DAgger changes the executing teacher/student across rounds. Training ends at 35 s by truncation or earlier upon physical failure; successful DAgger collections also stop early. Figure 3 reports optimization and stochastic collection outcomes, distinct from the frozen evaluations.

TABLE III

SPATIAL PROBES. EACH ROW IS ONE NOMINAL TRIAL; CONFIGURATIONS DIFFER.

Controller	Goal	Final roll	Outcome
Teacher	45°	44.29°	Pass
Teacher	90°	88.29°	Pass
Teacher (outside scope)	180°	51.40°	Tracking failure
Learned actor	15°	12.83°	Pass
Learned actor	45°	17.26°	Self-penetration

IV. SPATIAL REFERENCE AND PROBE PROVENANCE

The spatial imitation actor uses 54 collection episodes in three rounds, 60 fitting epochs per round, and planner-selected targets from $\{0°, 15°, 45°\}$. Initialization transfers the upper/lower-grasp planar actor. After the first round, the configured teacher fraction is 0.25. The gallery-panel mass is fixed at 1.5 kg; activation is synchronized, without asynchronous early lift. Pose observations append local position/rotation error, desired and measured spatial twist, contact geometry and stance error. The actor still outputs 21 commands, with bounded arm/lift rates during spatial tracking. Pose references and phase information remain shared in training and execution. The 90° demonstration is geometric-teacher control, not this trained actor. These probes receive no PPO update; subsequent spatial PPO runs are separate development studies.

The illustrated teacher probes use upper/lower rail spacing 0.20 m; the learned probes use 0.16 m. A 0.15 m clearance lift takes 8 s after a 2 s post-lift hold. Angular interpolation takes 22.5 s for 45° and 45 s for 90°. The generic historical reference generator also permits 180° commands; the manuscript's 90° fixed-grasp envelope is a declared scope restriction, not a retrospectively applied termination guard. Roll is relative to the initial lifted object frame. No large-roll probe establishes doorway passage or all-pose reachability.

The complete same-start learned test contains 0°, 15° and 45° targets: two of three pass. The five website illustrations above are selected probes. The adaptive teacher index retains all 42 attempts (six passing), with changing geometry and controller settings; neither that count nor the selected successes estimates frozen-controller reliability. The failed 180° trial stops before 90° and therefore cannot identify 90° as an experimentally measured failure threshold.

V. LATER PLANAR COHORT

The twelve-scene evaluation fixes the actor checkpoint and declares 48 starts before execution. It uses 1–2 kg loads, $\pm 5\%$ dimensions, 0.08–0.50 m base offsets, $\pm 20°$ headings, 0.10 m object-position jitter and up to 8 s activation delay. The budget is 180 s and planned clearance is 20 mm. Upper/lower handles extend panel acquisition; supervised progression and idealized weld retention remain. Its different actor and source configuration prevent attribution of the difference from 35/40 to a single modification. The two acquisition timeouts occur in the living-room painting and warehouse panel scenes. Every declared trial remains in the denominator. This cohort has no large-roll task and does

TABLE IV

LATER FROZEN PLANAR COHORT: ALL TWELVE SCENES.

Scene	Completed / declared
Apartment	4/4
Bedroom	4/4
Open arena	4/4
Gallery panel	4/4
Living room	4/4
Living-room painting	3/4
Loading dock	4/4
Office table	4/4
Dining room	4/4
Warehouse crate	4/4
Warehouse panel	3/4
Workshop beam	4/4
Overall	46/48

TABLE V

COMPLETED LOCAL-HISTORY DEVELOPMENT SUITES. COLUMNS ARE DISTINCT TASKS, NOT A POOLED SCORE.

Variant	Transport	Wait	Early release	Motion
V44	12/24	4/8	6/8	14/14
V45	12/24	7/8	7/8	12/14
V46	12/24	3/8	7/8	13/14
V47	0/24	0/8	0/8	0/14

not replace the original formation/clearance analysis. Actor hash, source manifest and per-scene outcomes are retained in `spatial_evidence.json`.

VI. LOCAL-HISTORY COORDINATION: IMPLEMENTED DEVELOPMENT TRIALS

A separate controller uses a shared causal transformer with private per-robot history, breakable finite attachments and local measurement ages. Its actor omits partner state, coordination phase and injected-fault flags. The spatial reference follows object progress; this differs from the timed spatial probes. A compact statement of the training/execution split is

$$\begin{aligned} a_{i,t} &= \pi_\theta(o_{i,t-H+1:t}), \qquad V_\psi(s_t^{\text{priv}}), \\ s_t^{\text{priv}} &= [q_t, 0.1\dot{q}_t, b_{1,t}, b_{2,t}, f_t]. \end{aligned} \tag{3}$$

where $q_t, \dot{q}_t$ contain both robots and the object, b_i are attachment indicators and f_t indicates fault injection. The centralized critic is used only in PPO training. Imitation centrally pools teacher-labeled robot samples; PPO pools rollouts and returns. Neither labels, partner histories, critic inputs nor optimizer state are supplied to the executing actors. Shared task references and simulated object estimates are still information inputs; no network budget is measured.

These four consecutive completed development variants are reported together rather than selecting the best score. V44 uses mixed-state imitation; V45 adds a current-observation head; V46 applies PPO; V47 varies release progress during collection. They are different training variants, not independent training seeds or a communication ablation.

The full suite contains 24 transport tasks and eight absent-partner waits. Transport comprises twelve nominal/delayed starts, eight later releases and four stale-pose cases. Later releases require four-hand support, object rise above 30 mm and path progress above 150 mm. Early-release and signed-motion suites have separate denominators. Safe waiting is not counted as transport completion. All variants reject payload/non-hand robot contact above 1 N; unsupported travel remains limited to 50 mm. V45 and V46 solve none of the eight later-release or four stale-pose cases. V47 fails all suites despite successful mixed-control collection, illustrating the gap between supervised fitting and closed-loop coordination. These results do not establish robust local recovery, broad scene transfer or learned 90° reorientation.

VII. FROZEN PLANAR FEEDBACK SEMANTICS

The frozen actor uses the legacy constraint feature: the first six equality solver multipliers for each palm weld are summed by row index, scaled by 0.01 and clipped to $[-5, 5]$ before observation normalization. The first three rows constrain translation and the next three orientation. This concatenation is not transformed into a calibrated robot-frame spatial wrench; in particular, the rotational rows cannot be interpreted directly as wrist torques. It is a simulator-specific feature, not validated force sensing. Removing or replacing it requires evaluation with the appropriate observation distribution.

Contact gates test acquisition, but active welds can sustain tensile forces and moments without a frictional retention limit. Recorded actuator models do not, by themselves, validate hand force capacity. The study reports neither actuator saturation distributions nor a load-to-capacity ratio; the 1–2 kg objects test large geometry rather than heavy-payload capability. Frictional slip and hardware sensor equivalence remain untested. The separate local-history suites inject loss of support under finite attachments; those results do not validate the original welded-retention model.

VIII. ADDITIONAL SPATIAL DIAGNOSTICS

IX. GEOMETRIC ILLUSTRATION

X. ADDITIONAL PLANAR DIAGNOSTICS

XI. NATIVE-CONTACT DEVELOPMENT BRANCH

An ongoing branch removes both grasp equalities and artificial hand–object spring wrenches. Articulated fingers transmit load through native MuJoCo contacts, with randomized high-friction materials and finite actuator limits. Contact-frame alignment, staged acquisition/lift/goal shaping and bounded useful-force credit draw on decentralized pinch transport, UniGraspTransformer [5] and GraspXL [6]. Additional costs measure tangential slip, excessive grip force, finger mechanical work and sustained torque. Opposing contact normals are a diagnostic proxy, not a force-closure proof. Fresh shared-transformer PPO has not established reliable native-contact transport. Fixture retention and initial failed grasp probes do not update any frozen manuscript cohort. Doorway, spatial-obstacle, reorientation and actual grasp-loss recovery require new independent validation. An actuator audit found missing inheritance of hand motor defaults in the initial native-contact experiments. The corrected model explicitly sets finger and wrist position gains to 2 N m/rad

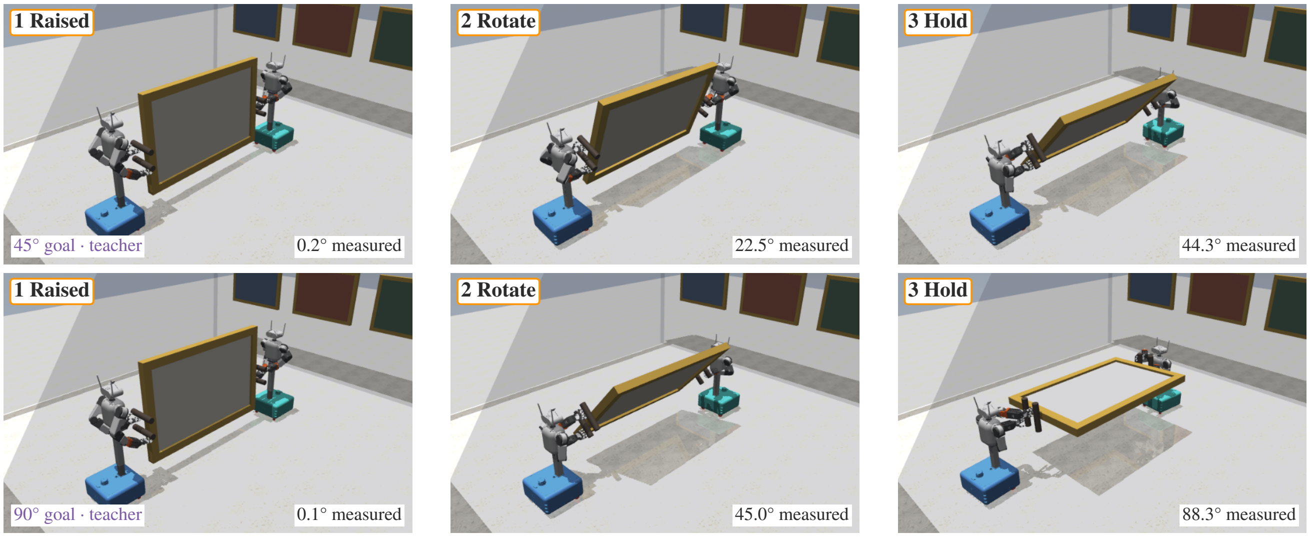

Fig. 4. Recorded fixed-grasp panel reorientation with the geometric teacher (seed 207000). Rows request 45° and 90° roll about the lifted object's long axis. Columns show clearance lift, intermediate rotation and terminal holding; labels give measured angles. Robot A/B have blue/teal bases and native body materials. Original joint states are rendered without stepping dynamics. These are pose-only probes, not learned-policy or doorway-transport successes.

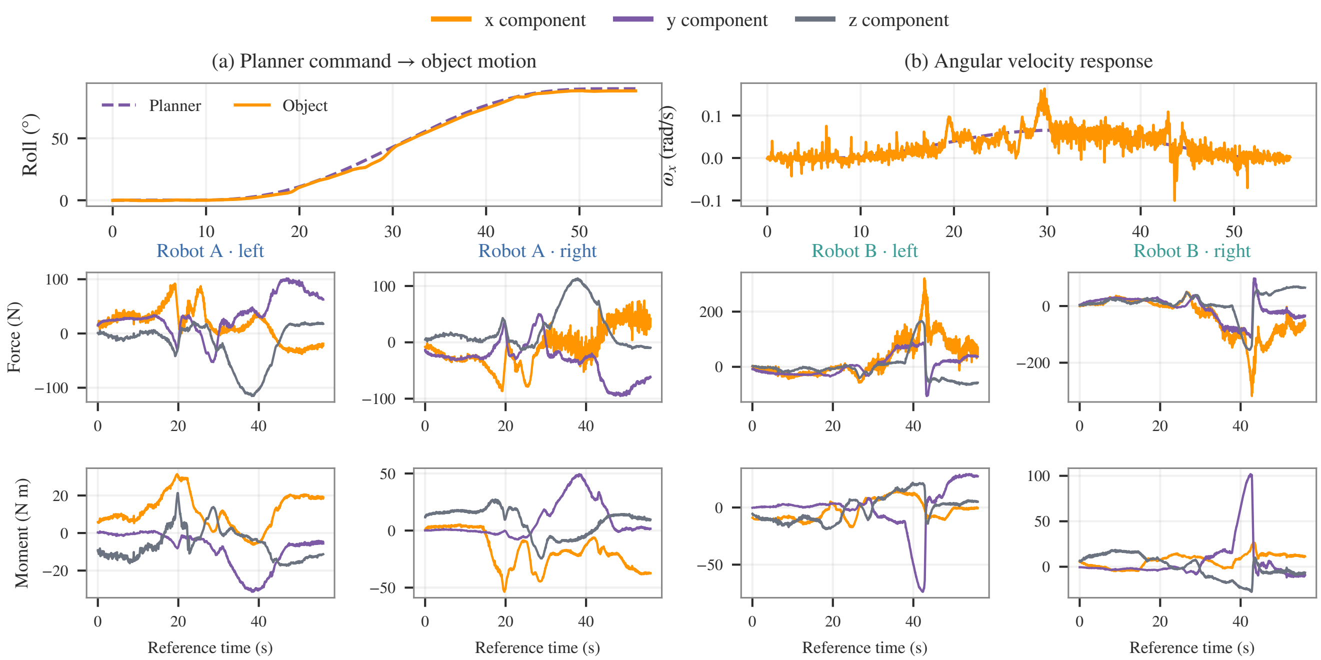

Fig. 5. Commanded object motion and resulting four-hand weld reactions in one instrumented 90° teacher probe. Dashed purple: pose/twist reference; orange: measured object response. Hand components use fixed lifted-object axes (x: orange, y: purple, z: gray); moments are palm-centered. Time starts at pose-schedule activation; ordinate scales vary by panel. These are simulated reactions, not commanded force targets or wrist-sensor data.

and torque limits to ± 1 N m. A scripted four-hand acquisition, 6 cm lift and 8 s hold succeeds without attachment forces. Repeating this diagnostic at 500 and 1000 Hz from identical initial states gives maximum object-trajectory differences of 1.25 mm and 0.16°. This is a short physics check, not learned transport evidence. A second supervised initialization has 10/12 successful teacher trials but 0/20 fresh frozen-actor acquisition successes; a third has 20/20 teacher successes but 0/20 fresh frozen-actor successes. Dataset aggregation collects corrective labels along perturbed student trajectories, with mixed-controller outcomes reported separately. A paired 32,768-step PPO comparison uses identical initial actor weights, training seeds and 20 evaluation seeds. A distance and outward-speed cost variant reaches four-hand support in 18/20 trials but completes 0/20 lift-and-hold tasks: 17 terminate on unintended payload–robot contact, two on self-contact and one on unsupported object travel. Adding a missing-secure-contact cost to that variant yields no four-hand acquisitions and 0/20 task successes. These single-training-seed development results do not establish a general reward ranking. A separate experiment adds decaying geometric action-target regression to PPO on actor-generated rollout

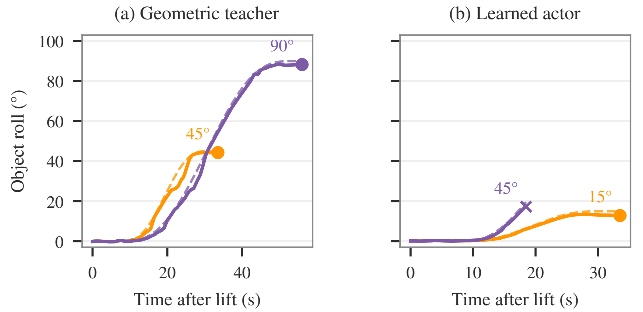

Fig. 6. Measured roll (solid) and evolving reference (dashed), relative to the lifted object frame. Circles mark successful completion; the cross marks self-penetration termination. Labels specify commanded endpoints, which are distinct from the evolving reference. Teacher and actor configurations differ.

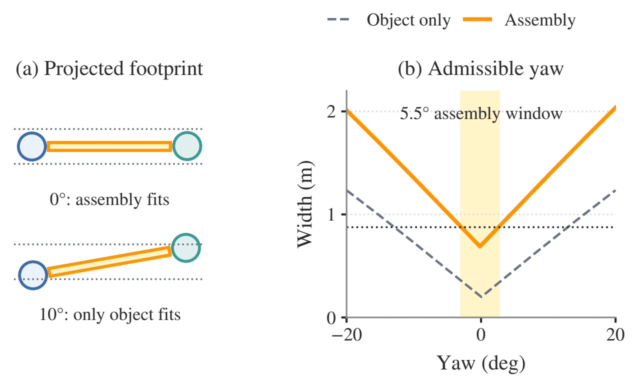

Fig. 7. Why the grasp changes passage feasibility. Analytic projections use the measured beam handoff geometry (seed 142024). (a) Dashed bounds show the available opening width after the 22 mm search margin on each side; the circles are base footprints. (b) The object-only yaw range is about 25.8°, versus 5.5° for the assembly (shaded); the horizontal dotted line is the door budget.

states, motivated by demonstration-assisted policy gradients [7]. The optimizer alone uses these labels; deployment remains the learned actor. This is not a reproduction of DAPG, and reliability has not been established. The completed baseline also yields 0/20 task successes and reaches four-hand support in 2/20 trials. The corrected-physics scene screen completes 0/26 doorway, obstacle, orientation and recovery cases. A bounded arm-position initialization has 24/24 teacher successes but 0/20 frozen-actor successes. A subsequent single-seed action-channel diagnostic replaces base, upper-body, gripper or all actor commands with geometric teacher commands. Only the all-teacher conditions succeed; hybrid outcomes do not count as learned results. The teacher's closing hysteresis was found to reinforce premature squeezing at approximately 7 cm palm error. Corrective-label schema 5 reduces its continued-closing window to 5 cm, retaining the 2.3 cm initial seating criterion. New data collection increases action disturbances only after eight successful trials at the current difficulty. This changes supervised labels and collection, not deployed action gates or physical acceptance criteria. The resulting learned policy still requires independent

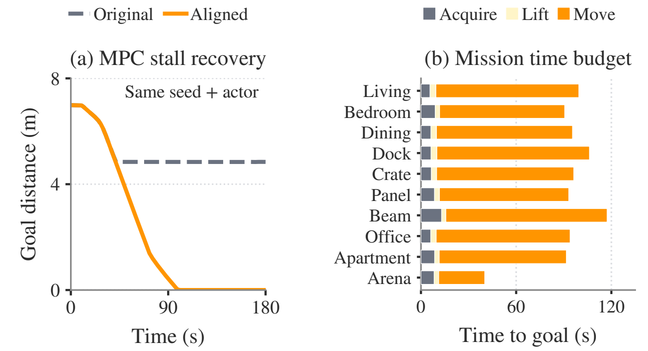

Fig. 8. Planning consistency and mission duration. (a) Same-seed, same-actor MPC comparison (114000): the original model stalls before the doorway; aligned geometry and margin permit progress to the goal. (b) Mean phase durations among successful frozen missions in each scene. Navigation dominates elapsed simulation time. The paired development case and frozen test use different actors and domains.

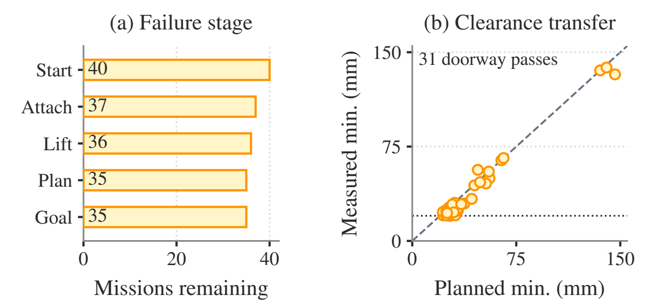

Fig. 9. (a) Mission counts at each stage, retaining all 40 starts. (b) Planned versus measured minimum clearance for 31 successful doorway missions; four arena passes have no finite clearance. Diagonal: equality; horizontal dotted line: 20 mm planning requirement.

evaluation. No native-contact experiment has established the requested full-mission reliability threshold.

XII. Reproduction Record

The accompanying `data/ral` records retain source hashes, imported-data provenance, reward definitions, actor/configuration identities and original outcomes. A review-response matrix distinguishes completed editorial and post-hoc analyses from missing controlled experiments. The spatial supplement freezes the site-linked probe sources and completed V44–V47 recovery suites; ongoing trials are excluded. The full-mission actor uses imitation; the reward table above is specific to this separate PPO study.

References

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